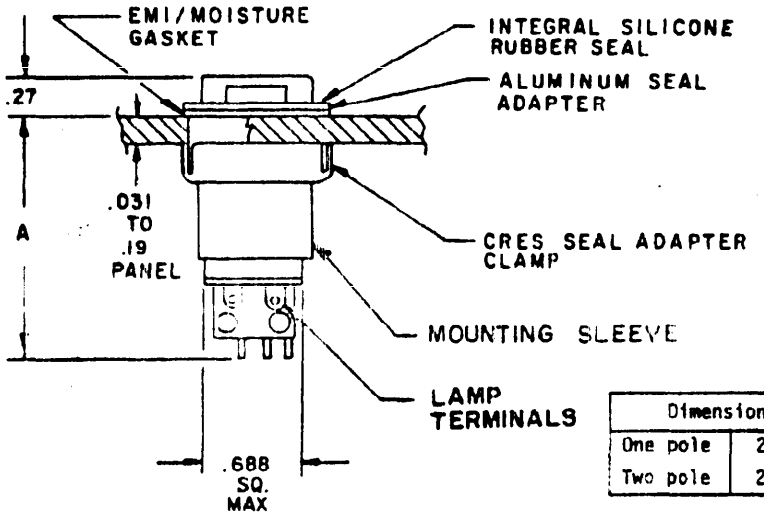
MILITARY SPECIFICATION SHEET

SWITCHES, PUSHBUTTON, ILLUMINATED MOMENTARY ACTION, 4-LAMP, SPDT AND DPDT, 7.5 AMPERES SILVER CONTACTS, 1 AMPERE GOLD CONTACTS (SPLASHPROOF, SUNLIGHT READABLE, HIGH IMPACT SHOCK RESISTANT, EMI SHIELDED)

This specification sheet is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

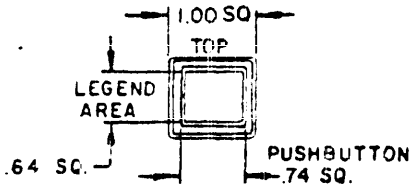
The complete requirements for procuring the switches described herein shall consist of this document and the latest issue of Specification MIL-S-22885.



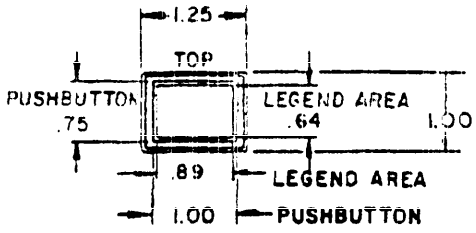
INCHES	MM
.005	.13
.03	.76
.031	.79
.19	4.83
.27	6.86
.64	16.3
.688	17.5
.74	18.8
.75	19.1
.89	22.6
1.00	25.4
1.25	31.7
2.11	53.6
2.16	54.9

Dimension A	
One pole	2.11 Max
Two pole	2.16 Max

BUTTON STYLE B



BUTTON STYLE D



Ⓐ

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 for three place decimals and ±.03 for two place decimals.
3. Basic switches shall be MIL-S-8805/101. Switch contact area sealed with silicone rubber sealant.
4. Pushbutton shall be designed to prevent incorrect insertion into switch housing.
5. Pushbuttons shall be held captive to switch body by retaining element to prevent accidental interchange.
6. Exact shape of switch optional provided dimensions specified are not exceeded.
7. Terminals and basic switch identification shall be permanently marked as shown.
8. Legend area for sunlight readable display is .480 square.

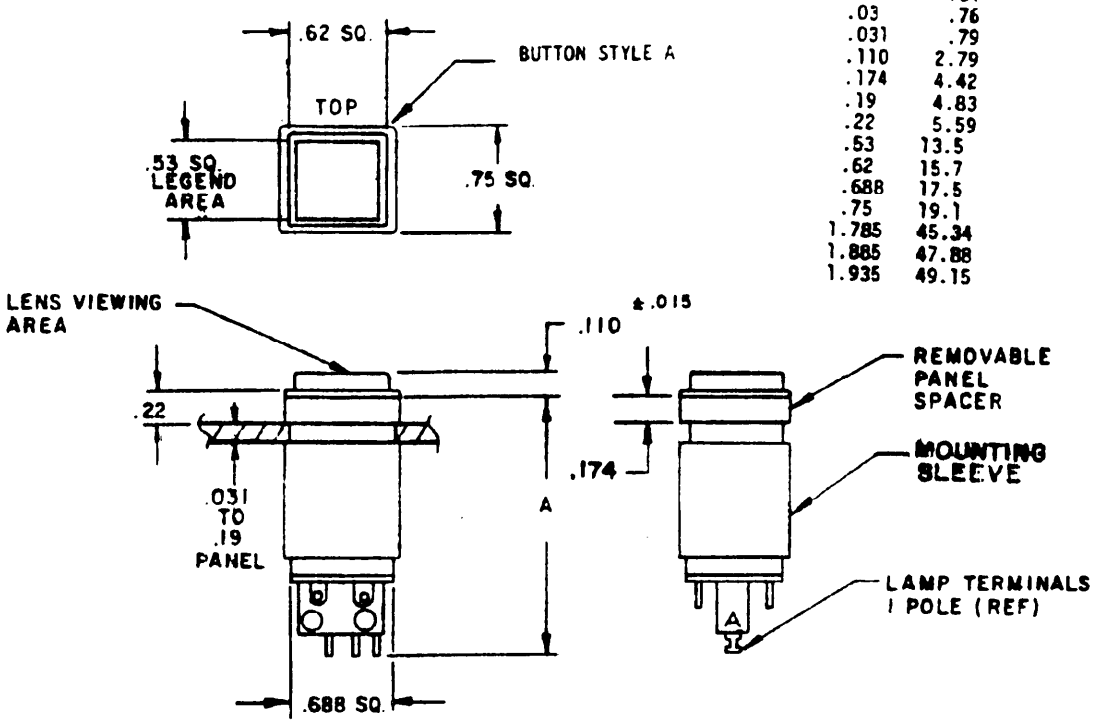
FIGURE 1. Switch, sealed or unsealed, button styles B and D.

Ⓐ Denotes changes.

FSC 5930

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INCHES	MM
.005	.13
.015	.31
.03	.76
.031	.79
.110	2.79
.174	4.42
.19	4.83
.22	5.59
.53	13.5
.62	15.7
.688	17.5
.75	19.1
1.785	45.34
1.885	47.88
1.935	49.15

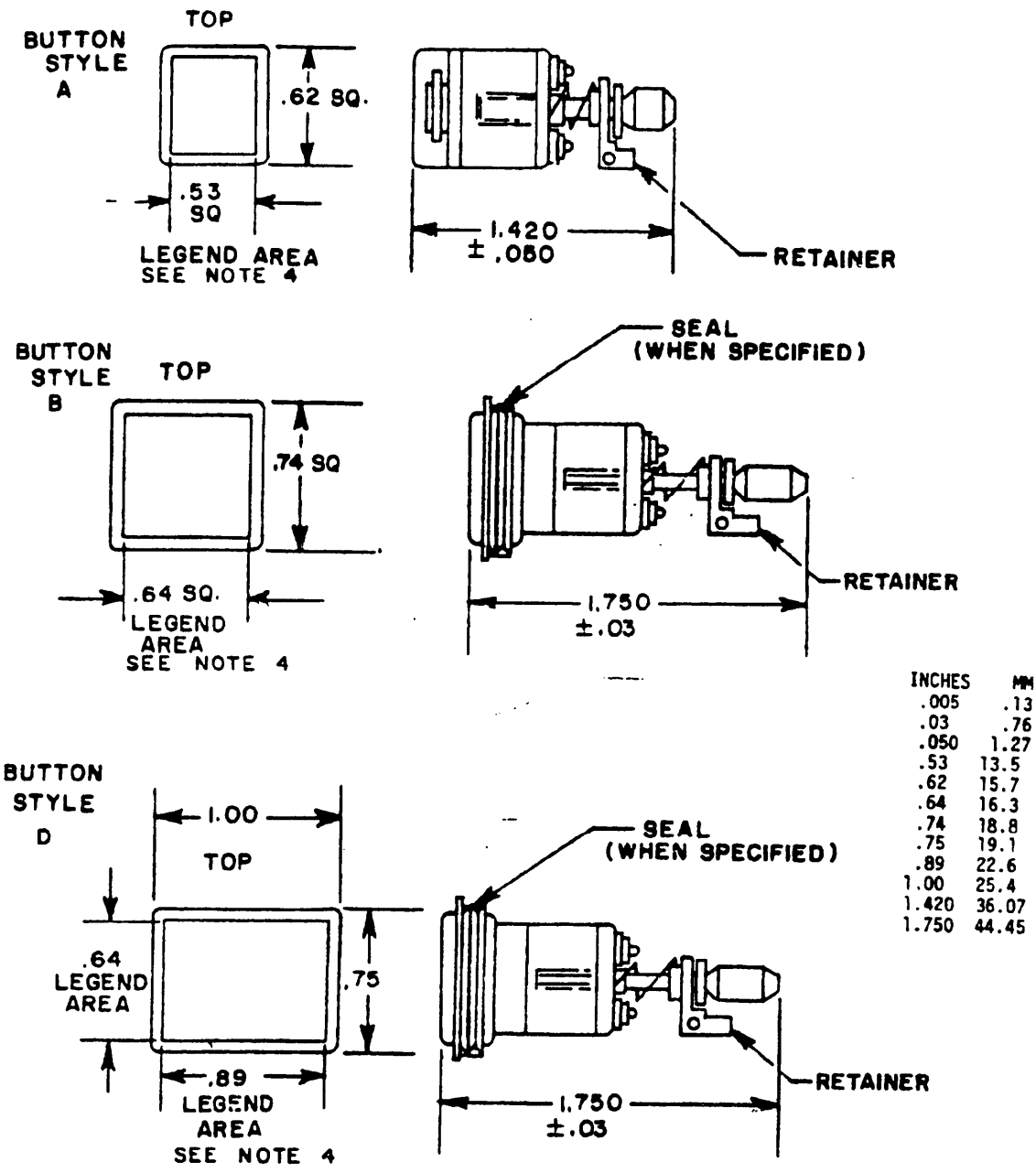


Dimension A	
One pole	1.885 max
Two pole	1.935 max
Indicator	1.785 max

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 for three place decimals and ±.03 for two place decimals.
3. Basic switches shall be MIL-S-8805/101. Switch contact area sealed with silicone rubber sealant.
4. Pushbutton shall be designed to prevent incorrect insertion into switch housing.
5. Pushbuttons shall be held captive to switch body by retaining element to prevent accidental interchange .
6. Exact shape of switch optional provided dimensions specified are not exceeded.
7. Terminals and basic switch identification shall be permanently marked as shown.
8. Legend area for sunlight readable display is .480 square.

FIGURE 2. Switch, unsealed, button style A.

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INCHES	MM
.005	.13
.03	.76
.050	1.27
.53	13.5
.62	15.7
.64	16.3
.74	18.8
.75	19.1
.89	22.6
1.00	25.4
1.420	36.07
1.750	44.45

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.005 for three place decimals and ±.03 for two place decimals.
3. Retainer holds pushbutton captive to switch body to prevent accidental interchange.
4. Legend area for all sunlight readable displays is .480 square.

FIGURE 3. Pushbutton styles.

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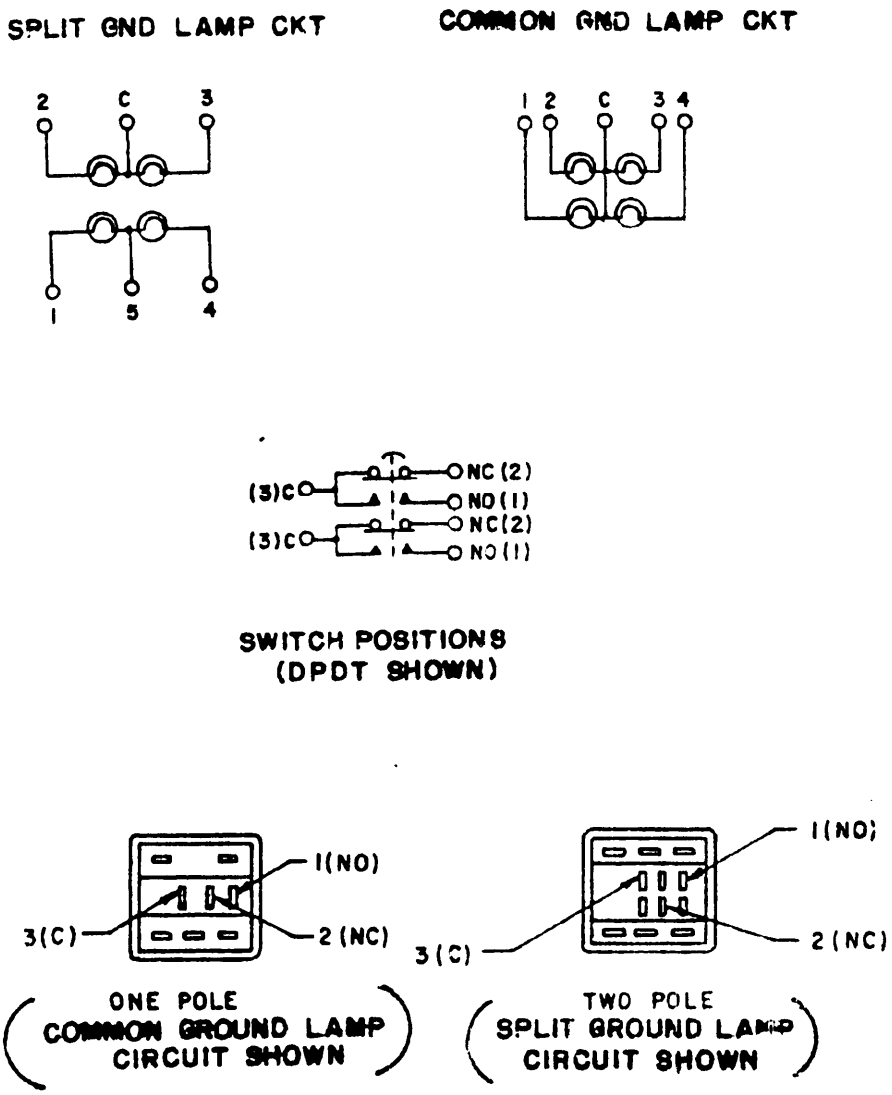
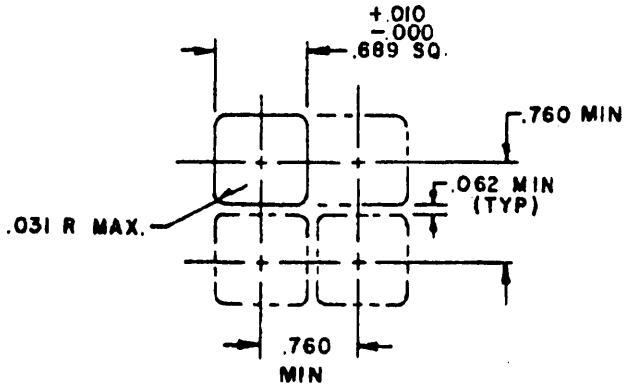


FIGURE 4. Circuitry and terminal identification.

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BUTTON STYLE

A

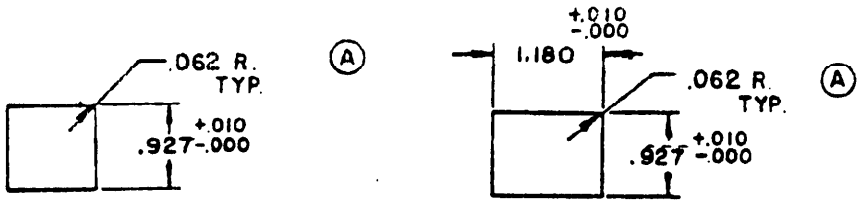


SEALED BUTTON

STYLE B

SEALED BUTTON

STYLE D



INCHES	MM
.010	.25
.031	.79
.062	1.57
.689	17.50
.760	19.30
.927	23.55
1.180	29.97

FIGURE 5. Recommended panel cutouts.

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REQUIREMENTS:

Design and construction: See figures 1 thru 4.

Enclosure design: Splashproof when specified, otherwise 1 (unsealed).

Temperature characteristic: 1 (-55° to +85°C).

Vibration grade: 3 (10-2000 Hz).

Lens transmittance: T (translucent) lens shall appear translucent white in the unlit condition. In the lit condition, it shall appear in a color determined by the color filters. Sunlight readable lens appear black in unlit condition. When lit, the legend appears in a color determined by the color filters.

Housing finish: Corrosion-resistant steel.

Operating characteristics:
Actuation force: 2.5 ±1.5 pounds sealed, 1.5 ±.5 pounds unsealed.
Actuation travel: .062 ±.025.

Lens extraction force: 1 to 4 pounds.

Shock: 75 G (MIL-STD-202, method 213, test condition H).

Thermal shock: During high temperature portion of thermal shock test all four lamps shall be energized with full rated voltage.

High impact shock: Applicable to sealed units only.

Splashproof test: Test in accordance with MIL-STD-108. There shall be no leakage of water through the panel and pushbutton seals as determined by visual examination and the dielectric withstanding voltage test.

EMI shielding requirements: When specified, switches shall be equipped with an EMI screen. When switches are tested in accordance with the following test method, the attenuation shall not be less than specified in figure 6.

Test method: Switches shall be tested to determine shielding effectiveness in accordance with the general procedures specified in MIL-STD-285. The test shall consist of electric field and plane wave measurements performed at the following frequencies:

Electric field	Minimum attenuation	Plane wave	Minimum attenuation
15 kHz	60 dB	100 MHz	47 dB
150 kHz	60 dB	400 MHz	39 dB
500 kHz	60 dB	1 GHz	33 dB
1 MHz	60 dB	2.5 GHz	28 dB
10 MHz	60 dB	10 GHz	20 dB

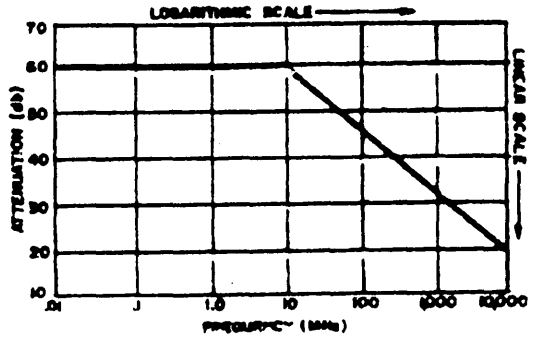


FIGURE 6. Minimum attenuation.

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Weight: 2 ounces maximum.

Electrical ratings: See table I.

(A) Low level life: Applicable for gold contact switches. 50,000 cycles.

Luminance: See table II.

Color: In accordance with MIL-S-22885/70. See table VI for sunlight readable displays.

QUALIFICATION:

All applicants for qualification approval shall demonstrate that each of their items conforms to all the requirements specified in the applicable documents, singularly and in combination with all the other previously qualified items, regardless of manufacturer.

Qualification inspection:

(A) Group submission: See table VII. Applies only when qualified MIL-S-8805/101 basic switches are utilized.

Group A inspection: Switches with sunlight readable legends shall be visually inspected under bright sunlight condition.

TABLE I. Electrical ratings. 1/

Contact finish	Load	Sea level		50,000 ft. 28 Vdc
		28 Vdc	115 Vac, 60 Hz	
		(amps, max)	(amps, max)	(amps, max)
Silver	Res.	7.5	5.0	4
	Ind.	5.0	3.5	2.5
	Lamp 2/	1.0	1.0	1.0
(A) Gold	Res.	1.0	---	1.0
	Ind.	0.5	---	0.5

1/ Minimum conditions are dependent on use and atmospheric conditions.
2/ Inrush, 10:1 for 20 milliseconds maximum.

Complete switch shall consist of:

1. Switch body.
2. Pushbutton with integral seal when specified.
3. Seal adapter, clamp and gasket (style B and D button only).
4. Mounting sleeve.
5. Edge light panel spacer (style A button only).
6. Pushbutton accepts 4 T-1 3/4 midget flange base lamps (not included, order separately). Switches with display code 0 (see table V) will accomodate lamps in the upper left and lower right positions only. Sunlight readable push button accepts 4, T-1 midget flange base lamps.

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TABLE II. Luminance (other than sunlight readable.

Color	Color code	Brightness (foot-lamberts) 1/ Without EMI screen 2/	With EMI screen 3/
Blue	B	8.0	4.0
Green	G	50.0	25.0
Red	R	30.0	15.0
White	W	180.0	90.0
Yellow	Y	300.0	150.0

1/ When illuminated by two MS25237 lamps of .30 ±.05 mean spherical candlepower at 28 volts dc. Readings are for display code 0 (see table V).
2/ Minimum average.
3/ For information only.

Part numbers: Part numbers are assigned as follows:

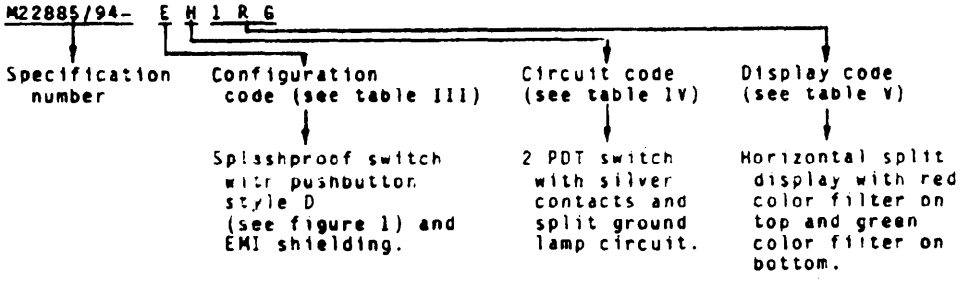


TABLE III. Configuration code.

Configuration code		Button style	Seal	EMI shielding
A		A (fig. 2)	No	No
B	1/	B (fig. 1)	No	No
C	1/	D (fig. 1)	No	No
D		B (fig. 1)	Yes	No
E		D (fig. 1)	Yes	No
F		A (fig. 2)	No	Yes
G	1/	B (fig. 1)	No	Yes
H	1/	D (fig. 1)	No	Yes
J		B (fig. 1)	Yes	Yes
K		D (fig. 1)	Yes	Yes

Ⓐ 1/ Inactive for new design.

TABLE IV. Circuit code.

Circuit code	Contact material	Circuit	Lamp circuit
A	gold	1 PDT	common ground
B	gold	2 PDT	common ground
C	gold	1 PDT	split ground
D	gold	2 PDT	split ground
E	silver	1 PDT	common ground
F	silver	2 PDT	common ground
G	silver	1 PDT	split ground
H	silver	2 PDT	split ground
J	-	indicator	common ground
K	-	indicator	split ground
L	-	none (push button only)	common ground
M	-	none (push button only)	split ground

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TABLE V. Display code. 1/

Display code 2/	Display
0X	
1XX	
2XX	
3XXXX	
4XXX	
5XXX	
6XXX	
7XXX	
8	None (switch body only)

1/ When legends are required, push buttons shall be procured separately from a source listed on the qualified products list.
2/ Codes shown are for push buttons without color filters. Replace X s with the color symbols of table II in the sequence shown.

Sunlight readable capability (with or without EMI shielding): This switch assembly may be obtained with legends that are visible when illuminated in direct sunlight. The legends are hidden when not illuminated. Illumination is by 4, 5 volt, T-1, .15 MSCP lamps. Other T-1 lamps are available and interchangeable with different illumination results. Available area for legend illumination in all push button styles is .48 x .48 inches. The illuminated chromaticity limits are as specified in table VI. When specified in the Qualified Products List, these switches have been tested to and passed the sunlight readable legend criteria as noted below.

Legend visibility-bright sunlight condition: All legends shall be capable of being read in direct sunlight. The average contrast ratio of any lighted legend to the background shall be 0.6 minimum when the display is subjected to ambient illumination of 10,000 foot candles minimum, at 5,000 $\pm 500^{\circ}$ Kelvin color temperature, directed at an angle of $45^{\circ} \pm 2^{\circ}$ to the normal of the viewing surface. The contrast ratio of any unlighted legend shall be less than 0.1 when the display is subjected to the above illumination conditions.

The ON/BACKGROUND contrast is defined by $C_L = (B_2 - B_1)/B_1$

The OFF/BACKGROUND contrast is defined by $C_{UL} = \left|(B_3 - B_1)/B_1\right|$

B_1 = Background luminance.
B_2 = Legend luminance, legend illuminated.
B_3 = Legend luminance, legend not illuminated.

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Luminance shall be measured at three points on each legend character. The average contrast of each character shall be reported.

TABLE VI. Illuminated color limits - sunlight readable displays. 1/

	X	Y
Red	.695 .710 .655 .660	.285 SL 2/ .325 SL 2/
Green	.300 .300 .380 .380	SL 2/ .600 .600 SL 2/
Yellow	.562 .570 .596 .606	.415 SL 2/ .382 SL 2/
Blue	.230 .230 .320 .320	.420 .350 .350 .420
White	.400 .460 .400 .460	.420 .420 .380 .380

1/ The chromaticities of the colors are expressed as "X" and "Y" coordinates on the CIE chromaticity diagram and are within the limits bound by the coordinates listed for each color.
2/ SL-spectrum locus (where intersected by other coordinate pair).

Ⓐ TABLE VII. Qualification inspection - group submission.

Switch	Qualification table of MIL-S-22885		Extent of approval
	Group	Number of switches	
M22885/94-ED3WWWW	I	20	All
	II	8 (from group I) 1/	
	III	2 (from group I)	
	VI	8 (from group I) 2/	
	VII	2 (from group I) 3/	
M22885/94-AFO(X)	I	10	
	II	4 (from group I) 4/	
	VI	6 (from group I) 5/	
	VIII	10 (2 of each color)	
M22885/94-FFO(X) with test legend 6/ REMOTE ALT LOW	VIII 7/	10 (2 of each color)	

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1/ Splashproof seal test to be performed after operating characteristics test. High impact shock test is applicable.
2/ Intermediate current test not applicable. Splashproof seal test to be performed after operating characteristics test.
3/ Splashproof seal test to be performed after operating characteristics test.
4/ Test 115 Vac loads only.
5/ Test legend to be sunlight readable type. Character size and spacing shall be selected to fully utilize the available legend area.
6/ In addition to group VIII tests, each of the switches shall be tested for legend visibility-bright sunlight condition. Two of these switches shall then be tested for EMI shielding effectiveness.

Preparing activity:
Navy - EC

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(See Instructions – Reverse Side)

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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